BROADACRE / SOWING

Liquid inject



Liquid prescription blends for your crop and soil.

We specialise in liquid prescription blends. Start with your soil test and crop requirements, then work with Hybrid-Ag to formulate a liquid blend suited to your crop, soil and seeding system.

> Bring us your soil test.

FROM THE SOIL TEST TO THE SEEDER

01

Test the soil

Identify what the crop and soil require with your Hybrid-Ag agronomist.

02

Formulate the blend

Use the Nutrient budget to formulate a liquid prescription blend.

03

Plan the delivery

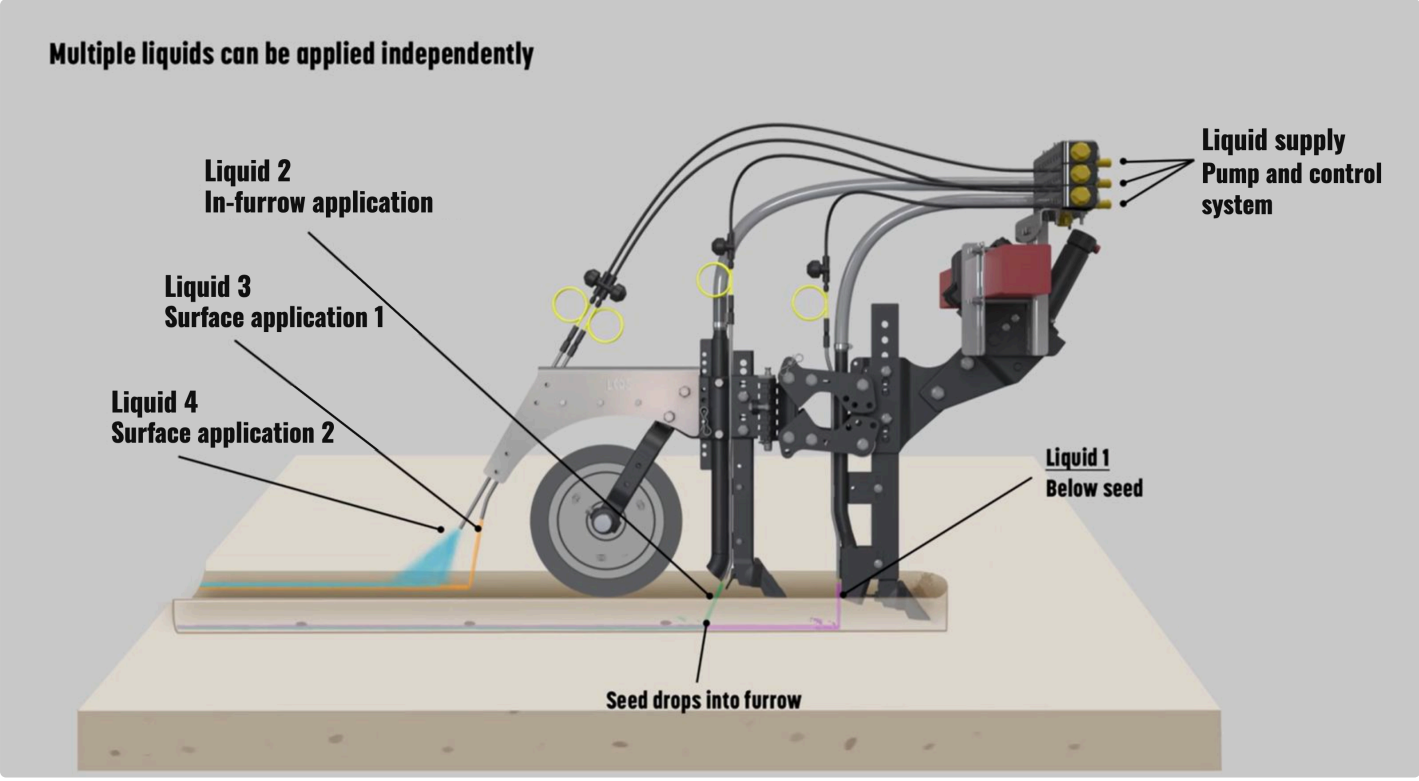
Agree the product rate, water ratio and placement before configuring the equipment.

Liquid injection can complement granular fertiliser. The Nutrient requirement still determines the overall inputs.

Placement makes the difference

Match liquid delivery to what the crop and soil require.

PLACEMENT OPTIONS AT SEEDING



Placement determines where Nutrients are delivered in relation to the seed and developing roots. The Nutrient form, concentration and soil moisture guide the position and separation needed for the chosen product.

Independent lines allow the delivery position and rate of each liquid to be specified separately.

NUTRIENT REQUIREMENTS TO CONSIDER

Nitrogen
Crop demand and seasonal supply.

Phosphorus
Soil supply and access for roots.

Calcium
Soil status and crop requirement.

Sulphur
The shortfall identified by testing.

Trace Elements
Assess each Element individually.

We specialise in liquid prescription blends. Your agronomist uses the crop and soil requirements to decide which Nutrients belong in the blend and what is suitable to deliver at seeding.

Illustrated positions are options, not a recommendation to apply every liquid or place every product with seed.

Everything starts with a soil test

Use the soil results to formulate the prescription.

We assess cation exchange capacity and base saturation alongside the full soil analysis, crop requirements and existing inputs. This helps establish the Nutrient quantities to plan per hectare.

Soil analysis

ANONYMISED EXAMPLE / 0-10 cm SAMPLE

CEC	TEC	PH (WATER)	ORGANIC CARBON
4.12	5.38	5.90	1.13%

Cation exchange capacity describes the soil's capacity to hold positively charged Nutrients. Base saturation describes the share of exchange sites occupied by each cation.

Cation balance

Base saturation	Found	Report target
Calcium	**50.00%**	64%
Magnesium	**11.52%**	15%
Potassium	**10.41%**	4-7%
Sodium	**1.49%**	0.5-1.5%

Selected base-saturation results and targets from this example report.

Requirements per hectare

kg/ha of each Element

Ca	Mg	S
Calcium	Magnesium	Sulphur
378	58	67

B	Zn	Cu
Boron	Zinc	Copper
4	16	10

These are the example report's total soil amendment requirements, rounded to whole kilograms. Product rates, timing and placement are planned separately.

From the soil result to the prescription

We use the Nutrient budget to **formulate a liquid prescription blend**. Longer-term soil correction is planned separately from crop nutrition, including what can be delivered at sowing.



Your paddock's results determine the recommendation.

Condensed example only. Report targets are specific to this example; these figures are not liquid-injection application rates.

Nitrogen in the prescription

Set the Nitrogen budget from crop demand, soil supply and existing inputs.

ToNik

LIQUID NITROGEN

ToNik supplies Nitrogen for crop growth. Its contribution is included in the overall Nitrogen budget.

APPLICATION

Agree the rate, water ratio and separation from seed with your Hybrid-Ag agronomist.



1,000 L IBC · illustration

PROVIDES Nitrogen

PRODUCT TYPE Liquid Nutrient product

Nutri-Core SA

LIQUID NITROGEN AND SULPHUR

Nutri-Core SA supplies Nitrogen and Sulphur in liquid form. Sulphur is needed for amino acid and protein synthesis.

APPLICATION

Use the soil and crop assessment to establish the requirement for each Nutrient before choosing the blend or standard product.



1,000 L IBC · illustration

PROVIDES Nitrogen and Sulphur

PRODUCT TYPE Liquid Nutrient product

Agree product suitability, rate, water ratio and placement with your Hybrid-Ag agronomist before sowing.

Building the prescription

Select each input for the requirement identified by the soil and crop assessment.

Nutri-Core NP

LIQUID NITROGEN AND PHOSPHORUS

Nutri-Core NP supplies Nitrogen and Phosphorus in liquid form. Phosphorus is needed for root development and crop growth.

APPLICATION

Use the soil results to set the Phosphorus requirement. Check compatibility before combining it with other products.



1,000 L IBC · illustration

PROVIDES Nitrogen and Phosphorus

PRODUCT TYPE Liquid Nutrient product

Liqui-Cal

LIQUID CALCIUM SUSPENSION

Liqui-Cal supplies Calcium in a fine Calcium carbonate suspension. Calcium is needed for new root growth and cell-wall development.

APPLICATION

Assess the crop and soil requirement. Suspension handling and any local pH effect need to be considered separately from the paddock's total lime requirement.



1,000 L IBC · illustration

PROVIDES Calcium 15.8% w/v

PRODUCT TYPE Nutrient suspension

Agree product suitability, rate, water ratio and placement with your Hybrid-Ag agronomist before sowing.

Targeted Nutrient inputs

Use the identified shortfall to guide the prescription or suitable standard product.

Liqui-Gyp

LIQUID CALCIUM AND SULPHUR SUSPENSION

Liqui-Gyp supplies Calcium and Sulphur in a fine suspension. The prescription accounts for the requirement for each Nutrient.

> **APPLICATION**
> Confirm product suitability, suspension handling and compatibility before liquid application.



1,000 L IBC · illustration

PROVIDES Calcium and Sulphur

PRODUCT TYPE Nutrient suspension

Opti-Trace Max

TRACE ELEMENT OPTION

Opti-Trace Max is a trace Element option to review against the crop and soil analysis. The required Elements and quantities determine the prescription.

> **APPLICATION**
> Confirm the current formulation and intended liquid application with your Hybrid-Ag agronomist.



1,000 L IBC · illustration

PROVIDES Trace Elements

SELECTION Based on testing

> *Agree product suitability, rate, water ratio and placement with your Hybrid-Ag agronomist before sowing.*

Prepare the delivery system

The blend and the equipment need to work together.

Once the prescription is agreed, use accurate product information to configure the equipment. Product properties and the intended application determine the setup.

Product information

Use current pH, specific gravity and product information when specifying the delivery system. A suspension may need different handling from a solution.

Mixing and handling

Check compatibility with the intended water source and follow the current product directions. Confirm agitation, filtration, storage and cleaning requirements before use.

Rate and placement

Set the application rate and product-to-water ratio with your agronomist. Check the concentration and separation needed to protect the seed.

Calibration

Calibrate for the actual product and planned ground speed. Check the delivery at each outlet before sowing.

> A jar test checks initial physical compatibility. It does not establish seed safety, crop response or financial return.

Keep the liquid application within the overall Nutrient budget. Adoption of a liquid system does not change the crop's Nutrient requirement.

Talk to us about your paddock

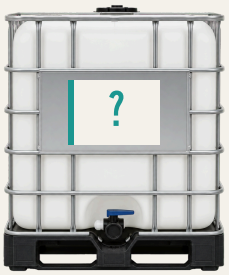
Bring your soil results and discuss a liquid prescription blend.

Tell us about the crop, the soil results and the inputs already planned. Your Hybrid-Ag agronomist can work through the Nutrient requirements and what is appropriate to deliver at seeding.

Start with your Hybrid-Ag agronomist

We specialise in liquid prescription blends. Standard Hybrid-Ag products may also be suitable where they meet the identified requirement.

hybridag.com.au



BROADACRE TEAM

James White and Bruce Armstrong can help you work through the crop and soil requirements.



James White



Bruce Armstrong

LIQUID DELIVERY AND EQUIPMENT

Hybrid-Ag supplies the agronomy, products and Nutrient prescription. Liquid Systems (SA) provides specialist advice on liquid delivery equipment and system configuration.

liquidsystems.com.au

Cover photography and the adapted placement diagram supplied by Liquid Systems (SA).

Product selection, application rate, mixing and placement require an agronomic recommendation.